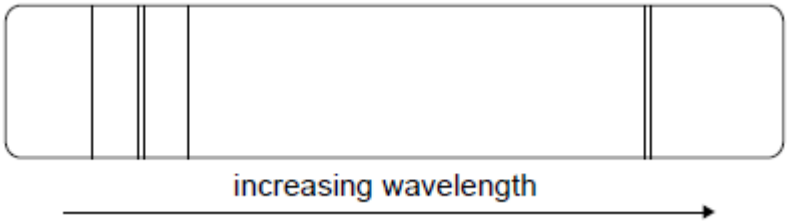
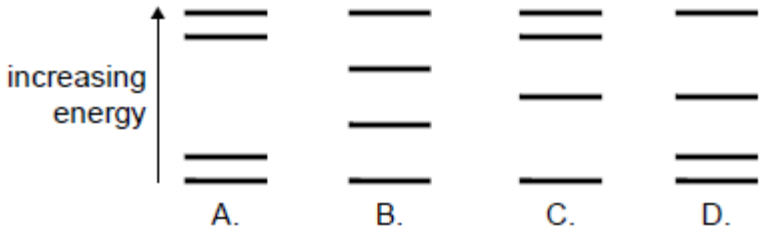
Paper questions E.1: Structure of the atom [31 marks]

1. [Maximum mark: 1]

The diagram shows the emission spectrum of an atom.



Which of the following atomic energy level models can produce this spectrum?

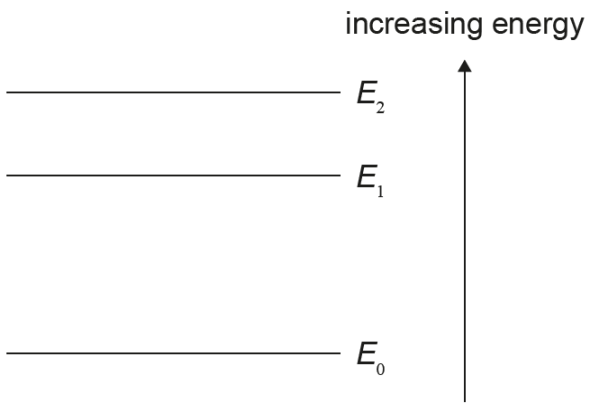


[1]

Markscheme
A

2. [Maximum mark: 1]

Three electron energy levels of a gas atom E_0, E_1 and E_2 are shown.



The following three claims are made about photons associated with these energy levels:

I. Transition from E_2 to E_1 will produce photons with the shortest wavelength.

II. Transition from E_2 to E_0 will produce photons with the highest frequency.

III. Three lines in the spectrum of this gas appear.

Which statements are **correct**?

[1]

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Markscheme
C

3. [Maximum mark: 8]

(a) Quarks are elementary particles.

(a.i) State what is meant by an elementary particle. [1]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

A particle without structure/constituents/component particles ✔

(a.ii) List the fundamental forces that act on quarks. [1]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

Electromagnetic, weak «nuclear», strong «nuclear» «and gravitational» ✔

Allow electroweak for electromagnetic and weak

(a.iii) The lambda baryon (Λ^0) has quark content uds. It decays according to the reaction $\Lambda^0 \rightarrow p + \pi^-$. The quark content of the pion is $\bar{u}d$.

State and explain which fundamental interaction is responsible for this decay. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

Weak ✔

Decay violates strangeness «and only the weak interaction does» ✔

For MP2 allow comments that suggest that strangeness has changed in the interaction

(b.i) By reference to the concept of binding energy, explain why the combined mass of the nucleons of a nucleus is greater than the mass of the nucleus. [2]

Markscheme

According to $E = \Delta mc^2$ / identifies mass energy equivalence ✔

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» ✔

(b.ii) The following binding energies per nucleon are available:

Th: 7.645074 MeV
Ra: 7.679917 MeV
He: 7.073915 MeV

Determine the energy released in the decay
$^{228}_{90}\mathrm{Th} \rightarrow {}^{224}_{88}\mathrm{Ra} + {}^{4}_{2}\mathrm{He}$.

[2]

Markscheme

$Q = 224 \times 7.679917 + 4 \times 7.073915 - 228 \times 7.645074$ ✔

$Q = 5.520$ «MeV» ✔

Allow 8.8×10^{-13} *J*

Award [1max] if answer is negative.

4. [Maximum mark: 4]

Over 60 elements, apart from hydrogen, have been identified in the composition of the Sun.

(a) Explain how elements present in the Sun are identified. [2]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

mention of spectrum/spectral lines ✔

lines/wavelengths/transitions are specific for an element ✔

OWTTE

(b) The temperature of the surface of the Sun is 5780 K. Calculate, in nm, λ_{max} for the spectrum of the Sun. [1]

Markscheme

502 «nm» ✔

Accept 502 or 500

5.02×10^{-7} scores [1] if unit (m) stated.

(c) Planets and comets orbit the Sun. Distinguish, with reference to their orbits, between planets and comets. [1]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

(most) orbits of planets are nearly circular/slightly elliptical

OR

orbits of comets highly/more elliptical ✔

OWTTE

5. [Maximum mark: 1]

What is the number of protons and what is the number of neutrons in a nucleus of $^{214}_{82}\mathrm{Pb}$?

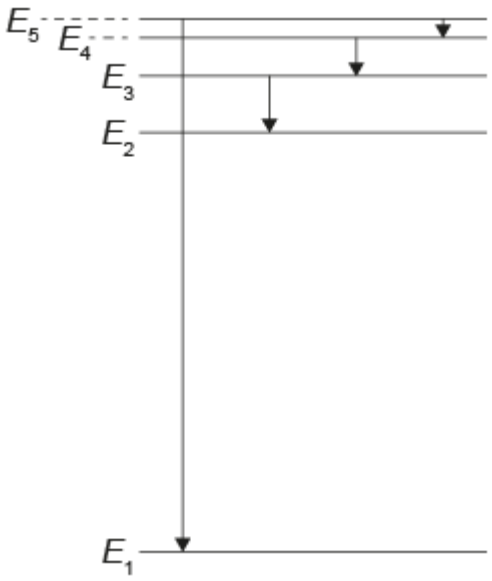
	Number of protons	Number of neutrons
A.	214	82
B.	82	214
C.	82	132
D.	132	214

[1]

Markscheme
C

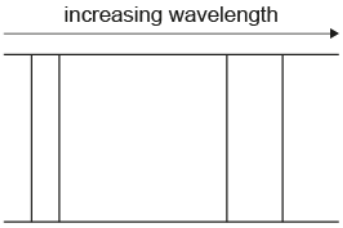
6. [Maximum mark: 1]

The energy levels *E* of an atom are shown.

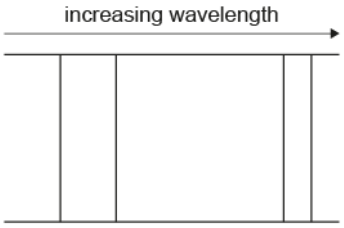


Which emission spectrum represents the transitions?

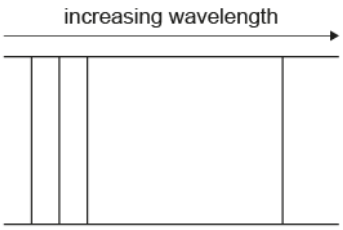
A.



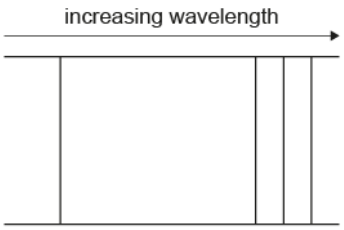
B.



C.



D.



[1]

Markscheme
D

7. [Maximum mark: 1]

Three claims are made about the structure of the atom.

I. Most of the atom is empty space.
II. The positive charge of the atom is concentrated in a small volume.
III. The electrons have discrete energy levels.

Which of these claims can be deduced from the Rutherford-Geiger-Marsden scattering experiment?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

[1]

Markscheme
A

8. [Maximum mark: 1]
Three claims are made about the structure of the atom.

I. Most of the atom is empty space.
II. The positive charge of the atom is concentrated in a small volume.
III. The electrons have discrete energy levels.

Which of these claims can be deduced from the Rutherford-Geiger-Marsden scattering experiment?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III [1]

Markscheme
A

9. [Maximum mark: 1]
White light is incident on gas at very low pressure. The light passing through the gas is analysed.

What phenomenon is being investigated?

A. Absorption spectrum

B. Emission spectrum

C. Photoelectric effect

D. Diffraction [1]

Markscheme
A

10. [Maximum mark: 1]
Three statements about the atom are:

I. The nucleus of the atom is positively charged.

II. The electrons provide only a small fraction of the mass of an atom.

III. Most of the atom is free space.

Which statements are correct?

A. I and II only

B. I and III only

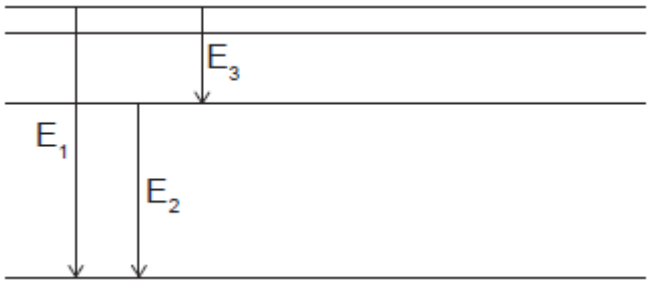
C. II and III only

D. I, II and III [1]

Markscheme
D

11. [Maximum mark: 1]

The diagram shows atomic transitions E_1, E_2 and E_3 when a particular atom changes its energy state. The wavelengths of the photons that correspond to these transitions are λ_1, λ_2 and λ_3.



What is correct for these wavelengths?

A. $\lambda_1 > \lambda_2 > \lambda_3$

B. $\lambda_1 = \lambda_2 + \lambda_3$

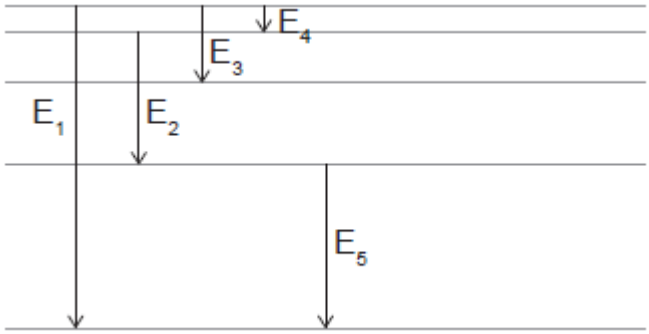
C. $\frac{1}{\lambda_1} = \frac{1}{\lambda_2 + \lambda_3}$

D. $\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$ [1]

Markscheme
D

12. [Maximum mark: 1]

Some transitions between the energy states of a particular atom are shown.



Energy transition E_3 gives rise to a photon of green light. Which transition will give rise to a photon of longer wavelength?

A. E_1

B. E_2

C. E_4

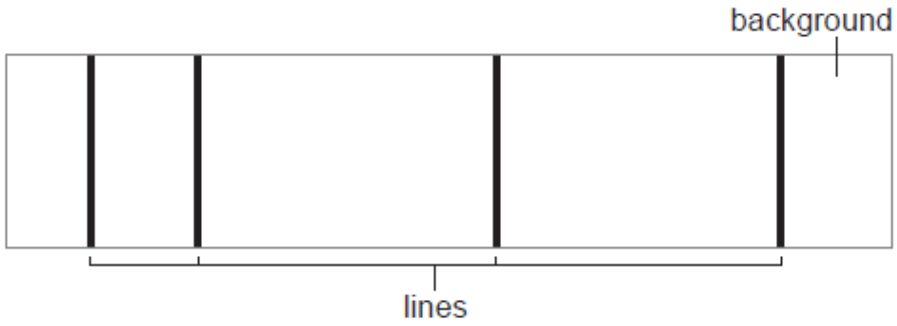
D. E_5

[1]

Markscheme
C

13. [Maximum mark: 1]

White light is emitted from a hot filament. The light passes through hydrogen gas at low pressure and then through a diffraction grating onto a screen. A pattern of lines against a background appears on the screen.



What is the appearance of the lines and background on the screen?

[1]

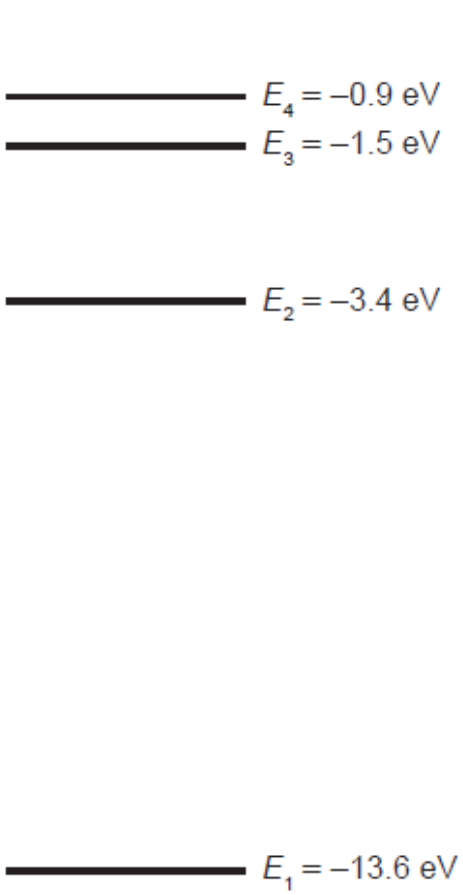
	Lines	Background
A.	dark	black
B.	white	coloured
C.	white	black
D.	dark	coloured

Markscheme

D

14. [Maximum mark: 1]

The energy levels of an atom are shown. How many photons of energy **greater** than 1.9 eV can be emitted by this atom?



[1]

A. 1

B. 2

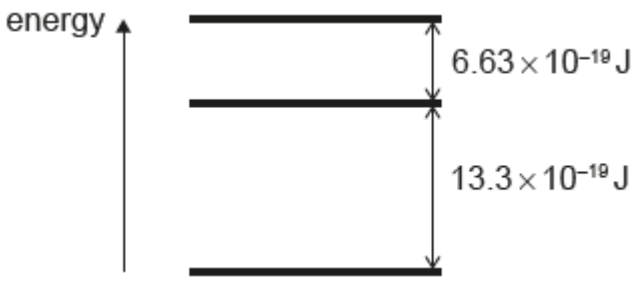
C. 3

D. 4

Markscheme
D

15. [Maximum mark: 1]

A simple model of an atom has three energy levels. The differences between adjacent energy levels are shown below.



What are the two smallest frequencies in the emission spectrum of this atom?

A. 0.5×10^{15} Hz and 1.0×10^{15} Hz

B. 0.5×10^{15} Hz and 1.5×10^{15} Hz

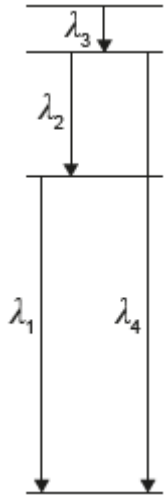
C. 1.0×10^{15} Hz and 2.0×10^{15} Hz

D. 1.0×10^{15} Hz and 3.0×10^{15} Hz [1]

Markscheme
C

16. [Maximum mark: 1]

The diagram below shows four energy levels for the atoms of a gas. The diagram is drawn to scale. The wavelengths of the photons emitted by the energy transitions between levels are shown.



What are the wavelengths of spectral lines, emitted by the gas, in order of decreasing frequency? [1]

A. $\lambda_3,\ \lambda_2,\ \lambda_1,\ \lambda_4$

B. $\lambda_4,\ \lambda_1,\ \lambda_2,\ \lambda_3$

C. $\lambda_4,\ \lambda_3,\ \lambda_2,\ \lambda_1$

D. $\lambda_4,\ \lambda_2,\ \lambda_1,\ \lambda_3$

Markscheme
B

17. [Maximum mark: 1]
The Rutherford-Geiger-Marsden experiment shows that

A. alpha particles do not obey Coulomb's law.

B. there is a fixed nuclear radius for each nucleus.

C. a large proportion of alpha particles are undeflected.

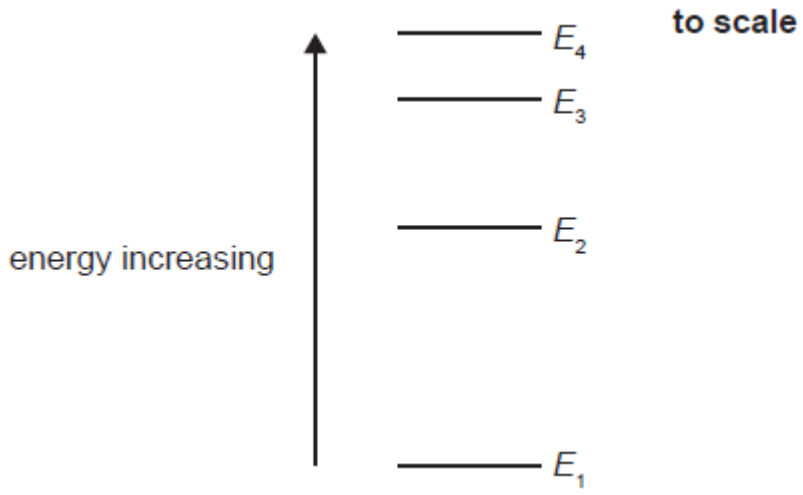
D. the Bohr model of the hydrogen atom is confirmed.

[1]

Markscheme
C

18. [Maximum mark: 1]

Four of the energy states for an atom are shown. Transition between any two states is possible.



What is the shortest wavelength of radiation that can be emitted from these four states?

A. $\frac{hc}{E_4 - E_1}$

B. $\frac{hc}{E_4} - \frac{hc}{E_1}$

C. $\frac{hc}{E_4 - E_3}$

D. $\frac{hc}{E_4} - \frac{hc}{E_3}$

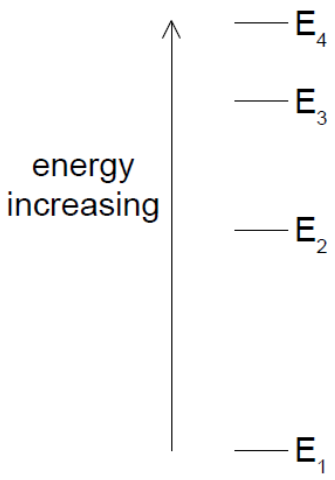
[1]

Markscheme

A

19. [Maximum mark: 1]

The energy levels for an atom are shown to scale.



A photon of wavelength λ is emitted because of a transition from E_3 to E_2. Which transition leads to the emission of a photon of longer wavelength?

A. E_4 to E_1

B. E_4 to E_3

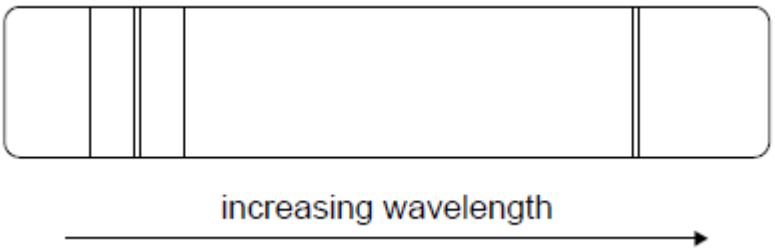
C. E_3 to E_1

D. E2 to E_1 [1]

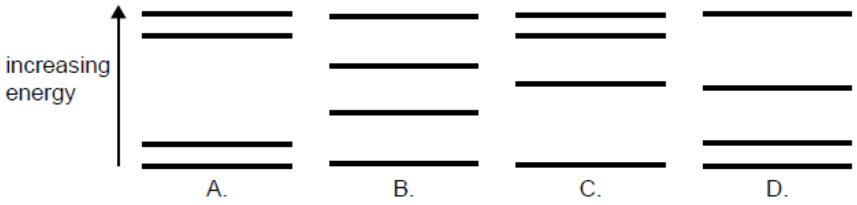
Markscheme
B

20. [Maximum mark: 1]

The diagram shows the emission spectrum of an atom.



Which of the following atomic energy level models can produce this spectrum?



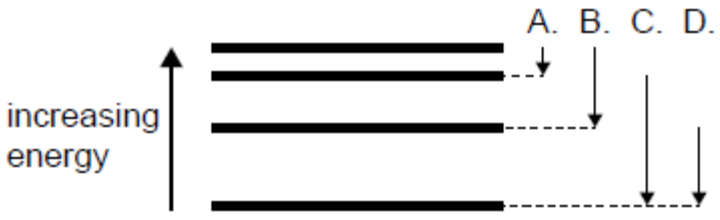
[1]

Markscheme

A

21. [Maximum mark: 1]

Which of the following atomic energy level transitions corresponds to photons of the shortest wavelength?



[1]

Markscheme

C

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